

Fahim Shahriyear Hossain

Senior Backend Engineer | Fintech & Payment Infrastructure | PCI DSS | Reliability Engineering

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 7+ years of experience building and scaling fintech and payment infrastructure. Specialized in backend architecture, distributed transaction processing, PCI DSS compliance, and reliability-focused platform engineering.

Experienced in modernizing legacy monolithic systems into scalable modular platforms using Clean Architecture and Hexagonal Architecture principles. Strong hands-on ownership across system design, backend engineering, deployment workflows, observability, and production reliability.

CORE EXPERTISE

Fintech Infrastructure • Payment Systems • Distributed Transaction Processing • PCI DSS Compliance • API Design • Backend Architecture • Reliability Engineering • System Modernization • High Availability Systems • Idempotency & State Management • CI/CD • Docker • Observability & Monitoring • Database Optimization • Team Leadership • Technical Mentoring • Server Patching

TECHNICAL STACK

Backend: PHP, Symfony, Laravel, Node.js, Express.js, REST API

Frontend: React.js, Next.js, Redux

Architecture: Clean Architecture, Hexagonal Architecture

Database & Storage: MySQL, Redis, Elasticsearch

Messaging & Async Processing: RabbitMQ

Cloud & DevOps: AWS (EC2, Route 53, S3, CodeDeploy), Docker, Jenkins, CI/CD, Linux, Bash Scripting

Observability & Monitoring: ELK Stack, Kibana, Centralized Logging, Monitoring & Alerting

Security & Compliance: PCI DSS, Tokenization, Encryption, Access Control, Audit Logging

Additional Exposure: Python, Go, MongoDB, N8N, LangChain

PROFESSIONAL EXPERIENCE

Portonics Limited — Dhaka, Bangladesh

Senior Software Engineer & Team Lead | March 2020 – Present

Led backend architecture, modernization, and reliability initiatives for PortPos payment infrastructure, focusing on scalable transaction processing, PCI DSS compliance, and platform resilience.

Key Contributions & Impact

- Led the modernization of a legacy monolithic payment platform into independently deployable systems, including payment page, merchant platform, administration platform, and unified invoice APIs.
- Managed backend systems handling approximately 300M BDT monthly transaction volume across 20+ payment gateways, serving 10M+ users and 500+ merchants.
- Designed the migration architecture using Clean Architecture and Hexagonal Architecture principles to improve maintainability, scalability, testing, and long-term framework flexibility.

- Integrated multiple payment methods, including Visa, Mastercard, American Express, Diners Club, bKash, Nagad, Rocket, Upay, and other local payment providers.
- Implemented state-machine-based transaction management and idempotency mechanisms to prevent duplicate payments and race conditions.
- Built scalable callback and IPN verification workflows to ensure reliable payment confirmation handling.
- Reduced transaction mismatch rate from 10% to 0.5% by redesigning transaction verification, reconciliation, and reliability workflows.
- Reduced payment failure and mismatch scenarios by 90% through improved retry strategies, transaction revalidation, and consistency handling.
- Improved overall platform latency by 60% while maintaining 99.99% uptime.
- Optimized database performance through indexing improvements, query optimization, transaction consistency improvements, and reduction of expensive join operations.
- Implemented centralized logging, monitoring, and observability using ELK Stack and Kibana.
- Built deployment pipelines using Jenkins, AWS CodeDeploy, Docker, and deployment scripting.
- Collaborated with DevOps teams to redesign infrastructure and deployment workflows during migration from monolith to modular systems.
- Led PCI DSS audit coordination with international QSAs, including documentation preparation, vulnerability assessment, penetration testing remediation, compliance discussions, and evidence handling.
- Led release coordination, code reviews, architecture discussions, technical mentoring, and engineering task planning for a 7-member team.
- Worked closely with project managers on sprint planning, engineering delivery, and production rollout coordination.
- Contributed to additional services, including recurring subscriptions, centralized notification systems, SendBill services, and backward-compatible invoice API migration.

Venus IT Limited — Dhaka, Bangladesh

Software Engineer | April 2019 – February 2020

Worked on ERP and financial management solutions for NGO operations, including KYC workflows, EMI modules, payment reminders, and reporting systems. Contributed to workflow automation and operational reporting improvements.

Link-Up Technology Limited — Dhaka, Bangladesh

Software Engineer | September 2018 – April 2019

Worked on retail POS and accounting solutions, focusing on reporting, billing workflows, and operational visibility improvements for retail businesses.

EDUCATION

- Bachelor of Science in Computer Science and Engineering
Daffodil International University, Dhaka, Bangladesh | 2019
- Diploma in Computer Technology
Naogaon Polytechnic Institute, Naogaon, Bangladesh | 2015

ADDITIONAL INFORMATION

Available for full-time remote opportunities with overlap for EU and US time zones.